

02YMECH - Homework 13

Assigned for the week of Dec 15, 2025

Question

Consider a rigid body with principal moments of inertia

$$I_1 < I_2 < I_3,$$

and body-fixed principal axes ($\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$). The motion is torque-free.

At time $t = 0$, the components of the angular velocity in the body frame are

$$\omega_1(0) = \omega_0, \quad \omega_2(0) = \omega_0, \quad \omega_3(0) = 0,$$

with $\omega_0 > 0$.

- (a) Write Euler's equations for this system and show that both the kinetic energy T and the squared angular momentum L^2 are conserved.
- (b) Using only these two conserved quantities, eliminate ω_2 and ω_3 to obtain a first-order differential equation for $\omega_1(t)$ of the form

$$\dot{\omega}_1^2 = f(\omega_1),$$

and determine the explicit polynomial f .

- (c) Determine the range of allowed values of $\omega_1(t)$ and show that the motion of ω_1 is periodic.
- (d) Express the period of $\omega_1(t)$ as an integral, clearly stating the change of variables used. Do not evaluate the integral.